REVIEW 1: PROJECT SETUP, SRS & WORKING APPLICATION DEMO

AegisPharm Enterprise

Smart Clinical Pharmacy Management System with Total Quality Management (TQM) Architecture

PROJECT PRESENTER:

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EVALUATION MILESTONE:

🏛️ **Amrapali University**

Evaluation: Review 1 — Setup, SRS & Working Application

Academic Year 2026

EVALUATION CRITERIA MAPPING

Review 1 Rubric Alignment & Verification Matrix

1. Setup & Baseline
Complete multi-tier project setup, dependency environment, and baseline clinical problem formulation.
📍 **Verified: Slide 3, 5 | `main.py`, `web_app.py`**

2. Modular Folder Architecture
Decoupled 6-tier directory layout separating core domain logic, UI bridges, API, and tests.
📍 **Verified: Slide 5 | `core/`, `ui/`, `apps/`, `tests/`**

3. Software Requirements (SRS)
Formal IEEE Std 830-1998 compliant SRS document covering functional & non-functional requirements.
📍 **Verified: Slides 7, 8, 9 | `SRS_AegisPharm_Enterprise.pdf`**

4. Scope & User Roles (RBAC)
5 distinct operational roles with workstation personas, terminal bindings, and route shields.
📍 **Verified: Slide 6 | `staff`, `active_sessions`**

5. Relational Database Design
14 normalized tables in SQLite with foreign keys, checks, default values, and ACID transactions.
📍 **Verified: Slide 10 | `core/db.py` (14 Tables)**

6. Implemented CRUD Operations
Comprehensive Create, Read, Update, Delete matrix verified across all 5 operational roles.
📍 **Verified: Slide 11 | SRS Section 6.2 & Live Code**

7. Working Application / Live Demo
Operational native desktop UI (Edge Chromium) and web portal (FastAPI) demonstrated live.
📍 **Verified: Slides 14, 15, 16 | `DEMO_GUIDE.md`**

8. Verification & Test Suite
24/24 passing automated tests verifying CDSS safety, FEFO logic, and Z-report math.
📍 **Verified: Slide 17 | `pytest` (24/24 Passed in 7.8s)**

RUBRIC: PROBLEM STATEMENT & BASELINE

Problem Statement & Baseline Clinical Understanding

🚨 1. MEDICATION DISPENSING ERRORS

• Clinical Reality: Over 1.3 million preventable patient injuries occur annually due to adverse drug events (ADEs).

• Handwriting & Transcription: Manual interpretation of handwritten prescriptions causes dosage miscalculations.

• Allergy Blindspots: Retail pharmacies lack real-time EHR cross-checks, risking fatal Penicillin or NSAID anaphylaxis.

• Baseline Remedy: Automated Clinical Decision Support System (CDSS) acting as an immutable quality shield.

❄️ 2. COLD-CHAIN & EXPIRY LOSS

• Biological Potency Loss: Vaccines, insulin, and biologics degrade silently outside the 2.0°C–8.0°C thermal envelope.

• FIFO Picking Flaw: Standard retail FIFO allows older batches to sit unnoticed while newer stock is sold.

• Dead-Stock Write-Offs: Significant annual capital lost in expired, unsellable inventory.

• Baseline Remedy: Algorithmic FEFO (First-Expiry, First-Out) picking + 24/7 continuous IoT temperature telemetry.

⚖️ 3. NARCOTIC COMPLIANCE & TILL DRIFT

• Schedule X / NDPS Violations: Inadequate tracking of habit-forming sedatives invites severe legal penalties and license revocation.

• Cash Drawer Shrinkage: Till variances, untracked mid-shift cash drops, and unverified digital UPI receipts.

• Failed Inspector Audits: Inability to generate instant batch reports during surprise CDSCO/FDA inspections.

• Baseline Remedy: Mandatory dual-PIN narcotic register + Daily Z-Report mathematical drawer reconciliation.

RUBRIC: QUALITY OBJECTIVE

Quality Objectives Aligned with TQM Principles

Objective 1: Zero Medication Dispensing Defects (Patient Focus)

Poka-Yoke error-proofing: Implement an automated Clinical Decision Support System (CDSS) that intercepts allergen cross-reactions (Penicillin, NSAIDs, Sulfa) and toxic drug-drug interactions before an invoice can be created.

Objective 2: 100% Cold-Chain Storage & Transport Compliance

Biological Integrity: Guarantee all thermo-sensitive pharmaceuticals are stored within the mandatory 2.0°C–8.0°C range via real-time 30s IoT telemetry streaming and automatic thermal spike alarms.

Objective 3: Zero Expiry Dead-Stock Losses (FEFO Scheduling)

Process Quality: Eliminate inventory wastage by algorithmically enforcing First-Expiry, First-Out (FEFO) batch prioritization, ensuring nearest-expiry viable medicines are dispensed first.

Objective 4: Zero Cash Register Discrepancy & Till Balancing

Fact-Based Management: Enforce strict shift-end financial governance through the Daily Z-Report engine, mathematically balancing opening float, sales, payouts, and counted physical currency.

Objective 5: 100% Regulatory Audit Readiness (ISO 9001 & CDSCO)

Continuous Assurance: Maintain an immutable SQLite audit trail, dual-PIN Schedule X narcotic register, and 1-click CSV export ready for State Drug Inspectors.

RUBRIC: MODULAR ARCHITECTURE

Modular Folder Architecture & 6-Tier Baseline

📁 ENTERPRISE FOLDER STRUCTURE

```
Pharmacy_Management_System/
├── core/
│   ├── bridge.py       # Unified Python RPC Bridge (38 methods)
│   ├── db.py           # SQLite persistence layer (14 tables)
│   ├── api_client.py   # Presence heartbeat AppConnector
│   └── logger.py       # Tamper-evident audit logging engine
├── config/
│   └── settings.py     # Thread-safe Singleton dataclass
(config.json)
├── ui/
│   ├── desktop/        # Edge Chromium WebView2 60 FPS frontend
│   └── web/            # Full-featured FastAPI Web UI (1:1 parity)
├── apps/               # Standalone admin & operations GUI apps
├── tests/              # 24 Automated unit & integration tests
├── backups/            # Scheduled .db snapshots & CSV audit exports
├── main.py             # Hardware-accelerated desktop entrypoint
└── web_app.py          # Asynchronous FastAPI web application server
```

🪜 6-TIER DECOUPLED ARCHITECTURE

Tier 1: Presentation Layer
Modern HTML5, responsive CSS3 Glassmorphism, and vanilla ES6+ JavaScript. Clean SPA design without external framework bloat.

Tier 2: Desktop Runtime
Microsoft Edge Chromium (WebView2) via `pywebview 5.x` providing zero-latency 60 FPS hardware acceleration.

Tier 3: Web Application Server
Asynchronous `FastAPI 0.138.0` + `Uvicorn 0.49.0` ASGI backend with auto-generated Swagger API documentation.

Tier 4: Unified Bridge RPC
`PharmacyBridge` exposes 38 secure, thread-safe Python methods directly to frontend JavaScript.

Tier 5: Core Domain Engines
Autonomous engines: CDSS allergy cross-matcher, FEFO dynamic batch sorter, cold-chain IoT simulator, Daily Z-Report engine.

Tier 6: Persistence & Storage
SQLite 3 ACID relational database with 14 normalized tables, foreign keys, and timestamped automated backups.

RUBRIC: SCOPE & USER ROLES

Project Scope & Granular User Roles (RBAC)

👑 Store Admin

Terminal: Console HQ

- Manage staff accounts & passwords
- Monitor live online terminal presence
- Review daily Z-Report reconciliations
- Execute SQLite snapshot backups
- Inspect system audit stream

🩺 Clinical Pharmacist

Terminal: Clinical Bay

- Review incoming e-Prescriptions queue
- CDSS allergy shield cross-checks
- Drug-drug interaction risk analysis
- Schedule X narcotic PIN sign-off
- Manage chronic patient refill cycles

💳 Cashier (POS)

Terminal: Billing Counter

- 1-Click verified eRx import to cart
- Auto FEFO nearest-expiry batch pick
- Split payments (Cash, UPI QR, Card)
- Calculate cash change return
- Print 80mm GST tax receipt

📦 Inventory Manager

Terminal: Supply Hub

- Multi-batch FEFO stock ledger
- Record Goods Receipt Notes (GRN)
- 24/7 Cold-chain IoT fridge monitor
- Manage supplier directory & POs
- Track low-stock alerts (≤20 units)

🛵 Delivery Agent

Terminal: Logistics Bay

- Active dispatch Kanban route board
- Verify cold-bag insulated packaging
- 4-Digit customer OTP verification
- Record electronic Proof-of-Delivery
- Handover timestamp audit log

RUBRIC: SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Software Requirements Specification (SRS) — IEEE 830

📜 IEEE STD 830-1998 SPECIFICATION STRUCTURE

Section 1: Introduction
Defines system purpose, document conventions, stakeholder reading guidelines, and regulatory references (CDSCO, GAMP 5, 21 CFR Part 11).

Section 2: Overall Description
Details system perspective, baseline problem formulation, 5 user classes with RBAC matrix, and operating environment constraints.

Section 3: System Features & FRs
Formal specification of 10 core Functional Requirements with exact input criteria, validation triggers, and expected outputs.

Section 4: External Interfaces
Specification of Hardware (USB HID scanners, 80mm ESC/POS printers, IoT probes) and Software (SQLite, FastAPI, WebView2) interfaces.

Section 5: Non-Functional Requirements
Quantifiable metrics for latency (<50ms), safety (Poka-Yoke CDSS blocking), security (SHA-256 RBAC), and 99.9% uptime.

Section 6: Database & CRUD Design
Normalized 14-table schema data dictionary, foreign key constraints, and comprehensive role-to-table CRUD permission matrix.

📁 SRS DOCUMENTATION DELIVERABLES

Formal Word Document (DOCX)
Generated via `generate_srs_doc.py` using `python-docx` with professional typography, custom slate headers, and styled tables: `SRS_AegisPharm_Enterprise.docx`.

Print-Ready PDF Specification
Compiled via Microsoft Word COM automation (`wdExportFormatPDF = 17`) ensuring zero layout degradation for university examiners: `SRS_AegisPharm_Enterprise.pdf`.

GitHub Markdown Version
Full specification available directly within repository root for instant web viewing: `SRS_AegisPharm_Enterprise.md`.

Traceability to Evaluation Rubric
Every rubric criterion from Image 1, 2, and 3 is mapped directly to corresponding functional sections and automated verification assertions.

RUBRIC: FUNCTIONAL SPECIFICATIONS

SRS Functional Requirements (FR-01 to FR-10)

📋 CORE CLINICAL & POS REQUIREMENTS (FR-01 TO FR-05)

FR-01: Authentication & Telemetry
Enforce SHA-256 hashed credentials, terminal session UUIDs, and 5s heartbeat telemetry. Offline flag after 15s inactivity.

FR-02: CDSS Clinical Safety Gate
Automated cross-check of eRx drugs against EHR patient allergies (Penicillin) and drug-drug matrices. Dispense lockout on conflict.

FR-03: Schedule X Narcotic Sign-Off
Mandatory dual authentication: Doctor DMC Registration Number + Pharmacist 4-digit Authorization PIN permanently logged.

FR-04: FEFO Batch Sorting Engine
Automatic prioritization of nearest-expiry viable lots (`days_to_expiry ASC`). Strict lockout of expired drugs (days <= 0).

FR-05: POS Multi-Tender Billing
1-Click import of approved eRx, barcode scan, Cash change calculator, Dynamic UPI QR generation, and 80mm GST receipt printing.

📋 SUPPLY CHAIN & FISCAL REQUIREMENTS (FR-06 TO FR-10)

FR-06: Goods Receipt Notes (GRN)
Inward supplier shipments against POs capturing lot number, expiry, cost price, MRP, GST %, and physical shelf rack coordinate.

FR-07: IoT Cold-Chain Telemetry
Monitor refrigerator sensors (Chamber A & B, 2°C–8°C) every 30s. Trigger audio/visual alert on temperature excursion spike.

FR-08: Delivery Logistics & OTP e-POD
Kanban dispatch workflow (`Packaging`, `Out`, `Delivered`), insulated cold-bag checklist, and 4-digit recipient OTP handshake.

FR-09: End-of-Day Z-Report Close
Shift cash reconciliation: `Expected = Float + Sales - Payouts`. Variance detection, register locking, and official signed Z-slip.

FR-10: 21 CFR Part 11 Audit Trail
Append-only, immutable SQLite event log capturing timestamp, user ID, role, action code, and severity level (INFO/WARN/CRIT).

RUBRIC: NON-FUNCTIONAL & QUALITY METRICS

SRS Non-Functional & Safety Requirements

⚡ 1. PERFORMANCE & LATENCY

SUB-SECOND RESPONSE STANDARDS

- UI Transaction Latency: All dashboard actions & button clicks respond in <50ms.
- Database Commit Latency: SQLite WAL transactions complete in <10ms.
- 60 FPS Desktop Rendering: Hardware-accelerated Edge Chromium WebView2 prevents UI stutter.
- Non-Blocking Threading: Zero frame freezes via asynchronous background worker threads.

🛡️ 2. POKA-YOKE SAFETY GATES

MISTAKE-PROOFING HEALTHCARE

- 100% CDSS Interception: Automatic lock on cart checkout when drug-allergy contraindication is flagged.
- Zero Expiry Dispensing: Expired batches mathematically prohibited from cart addition.
- Schedule X Dual-Auth: Narcotics cannot be billed without Doctor DMC Reg + Pharmacist PIN.
- Handover Integrity: Delivery status advancement blocked without 4-digit recipient OTP.

🔒 3. SECURITY & AUDIT TRAIL

21 CFR PART 11 COMPLIANCE

- Immutable Audit Records: Append-only SQLite event stream; UPDATE and DELETE disbarred.
- Role-Based Route Shields: Strict RBAC gating on all sensitive operations and UI modals.
- Credential Protection: SHA-256 hashed password verification across all login terminals.
- Session Heartbeat Watchdog: Stale or hijacked terminal sessions auto-terminated after 15s.

⚖️ 4. STATUTORY COMPLIANCE

INDIAN PHARMACEUTICAL STANDARDS

- Drugs & Cosmetics Act 1940: Mandatory compliance with Schedule H, H1, and Schedule X rules.
- Rule 65 Dispensing Records: Automated digital maintenance of prescription registers.
- GST Invoicing Compliance: Automated 5%, 12%, 18% CGST/SGST tax split on thermal receipts.
- GAMP 5 Category 4: Configured software validation standards with complete test traceability.

RUBRIC: DATABASE / SYSTEM DESIGN

Database Design & Relational Data Dictionary

🗄️ CORE RELATIONAL TABLES (PART 1)

1. staff (PRIMARY KEY: id)
Staff credentials, assigned RBAC role, username, branch, active status.

2. medicines (PRIMARY KEY: id)
Pharmaceutical catalog, generic name, batch, expiry, days left, stock, MRP.

3. prescriptions (PRIMARY KEY: rx_id)
Patient name, doctor, items JSON, CDSS status, clinical sign-off notes.

4. invoices (PRIMARY KEY: invoice_id)
Official tax invoices, cashier, payment method, subtotal, GST, total amount.

5. delivery_orders (PRIMARY KEY: order_id)
Logistics dispatch, address, customer phone, rider, status, 4-digit OTP.

6. patients (PRIMARY KEY: id)
Patient EHR, phone, chronic conditions, loyalty points, recorded allergies.

7. suppliers (PRIMARY KEY: id)
Distributor master, contact, phone, billing email, GSTIN, payment terms.

🗄️ COMPLIANCE & TELEMETRY TABLES (PART 2)

8. purchase_orders (PRIMARY KEY: po_id)
GRN shipment log, supplier, items JSON, total amount, inward status.

9. narcotic_register (PRIMARY KEY: id (AUTO))
Schedule X log: Doctor DMC Reg No, Pharmacist PIN, batch, quantity.

10. temperature_logs (PRIMARY KEY: id (AUTO))
24/7 cold-chain telemetry: Sensor ID, temperature (2-8°C), alert flag.

11. refill_schedules (PRIMARY KEY: id)
Chronic adherence: Patient, medicine, last dispensed, next due date.

12. active_sessions (PRIMARY KEY: session_id)
Live presence: User ID, role, terminal name, last heartbeat timestamp.

13. audit_logs (PRIMARY KEY: id (AUTO))
Immutable security log: User, role, action, details, severity level.

14. z_reports (PRIMARY KEY: z_report_id)
Daily closing: Opening float, cash sales, payouts, variance, signed status.

RUBRIC: CRUD OPERATIONS & RBAC

Role-Based CRUD Operations Matrix

COMPLETE CRUD IMPLEMENTATION ACROSS ALL CORE MODULES

Module / Entity	CREATE	READ	UPDATE	DELETE
Staff & Users	Add staff user modal (ID, Name, Role, Branch)	Staff credentials list & live online presence	Edit staff role, password, branch & status	Delete unauthorized staff user with audit log
Medicines Catalog	Add medicine modal (SKU, Category, Batch)	Catalog table with FEFO visual countdown badges	Adjust physical stock delta (+/- units modal)	Delete expired/discontinued medicine SKU
Clinical Prescriptions	Inward e-Prescription modal with medicine items	Prescriptions review queue with CDSS allergy scan	Update verification status (Approve / Reject)	Archive processed prescriptions into sales ledger
Narcotic Register	Add Schedule X dispensation with Doctor Reg No	Read narcotic audit ledger with search filter	Update verified status with 4-digit Pharmacist PIN	Restricted: Immutable audit records (CDSCO law)
Invoices & POS	Generate sales invoice with 5% CGST/SGST split	Sales ledger history with 1-click receipt modal	Update patient loyalty points upon purchase	Void invoice requires supervisor override
Deliveries & Logistics	Create home delivery order with customer phone	Dispatch Kanban board (Packaging/Out/Delivered)	Advance delivery status & verify 4-digit OTP	Cancel unassigned delivery dispatch task
Daily Z-Report Close	Generate official Z-Report ID (Z-YYYYMMDD-XXXX)	Read gross sales, cash total, digital collections	Update counted cash & calculate live variance	Permanent archive: Immutable signed registry

RUBRIC: MODULES & QUALITY ALIGNMENT

Core Modules Aligned with TQM Quality Goals

🩺 Clinical Decision Support (CDSS)

ALIGNED TQM GOAL: ZERO MEDICATION DISPENSING DEFECTS

- Real-time allergen cross-checks (Penicillin, NSAIDs, Sulfa).
- Molecule-molecule drug interaction checker.
- Double sign-off for Schedule X narcotics (Pharmacist PIN + Doctor Reg).
- Patient chronic disease and allergy EHR integration.

📦 FEFO Inventory & Cold-Chain IoT

ALIGNED TQM GOAL: ZERO BIOLOGICAL POTENCY LOSS & ZERO DEAD-STOCK

- Strict First-Expiry, First-Out batch sequencing algorithms.
- Visual countdown tags (<30d Critical, <90d Attention).
- 24/7 continuous IoT refrigerator telemetry (2.0°C–8.0°C).
- Goods Receipt Notes (GRN) dockside supplier verification.

💳 Cashier Express POS & GST Billing

ALIGNED TQM GOAL: SERVICE SPEED, TAX INTEGRITY & ZERO CASH LEAKAGE

- 1-Click verified eRx import from clinical queue.
- Multi-tender checkout: Cash change calculator, Dynamic UPI QR, Card.
- Automated statutory 5% GST calculation (2.5% CGST + 2.5% SGST).
- Instant 80mm thermal receipt printing and sales ledger.

🛵 Delivery Dispatch Fleet (e-POD)

ALIGNED TQM GOAL: 100% LAST-MILE HANDOVER FIDELITY

- Visual 3-column route Kanban board (Packaging, Out, Delivered).
- Cold-bag insulated packaging verification checklist.
- 4-Digit customer SMS OTP verification before order handoff.
- Automatic GPS arrival and handover timestamp logging.

RUBRIC: INITIAL UI & MAIN WORKFLOW

Initial UI & End-to-End System Workflow

STEP 1

Stage 1: Prescription Intake

- Doctor e-Prescription uploaded
- Patient EHR allergies scanned
- Enters pending review queue

STEP 2

Stage 2: Clinical CDSS Gate

- Pharmacist runs CDSS scan
- Allergy & interaction checks
- Schedule X PIN authenticated
- Status updated to 'Approved'

STEP 3

Stage 3: POS Express Billing

- 1-Click import to Cashier cart
- Auto FEFO batch picking
- Dynamic UPI QR / Cash change
- Thermal 80mm GST receipt printed

STEP 4

Stage 4: Cold-Chain & Fleet

- Cold-bag checklist verified
- Dispatched on Kanban route
- Customer gives 4-digit OTP
- Marked 'Delivered' (e-POD)

STEP 5

Stage 5: Daily Z-Report Close

- Cash drawer counted
- Variance calculated (₹0.00)
- Shift signed & closed in SQLite
- CSV exported for accounting

🎨 USER INTERFACE DESIGN & DUAL DEPLOYMENT PARITY

- Desktop UI: Rendered via Microsoft Edge Chromium WebView2 engine with hardware acceleration (60 FPS) and zero native bridge latency.
- Web UI: Hosted asynchronously via FastAPI (Uvicorn), providing 100% feature parity on Chrome, Safari, Edge, tablets, and mobile.
- Design System: Clean White/Light mode default with 1-click Dark Mode toggle, responsive side navigation, and interactive modals.

RUBRIC: WORKING APPLICATION / DEMO GUIDE

Working Application Demo Guide: Setup & Personas

🚀 APPLICATION LAUNCH COMMANDS

Step 1: Automated Test Suite (Examiner Proof)

Command: `pytest`
Verifies 24/24 unit & integration tests in ~7.8s with zero failures.

Step 2: Native Desktop Workstation Application

Command: `python main.py`
Boots hardware-accelerated Edge Chromium WebView2 window with default Light Mode and instant Dark toggle.

Step 3: Cloud-Ready Web Portal Application

Command: `python web_app.py`
Launches asynchronous FastAPI server at `http://127.0.0.1:8000` with Swagger documentation at `/docs`.

Step 4: Physical Terminal Binding

Select terminal station: `Front Counter A`, `Clinical Kiosk B`, `Inwarding Bay`, `Delivery Dock`.

🔑 WORKSTATION PERSONAS & CREDENTIALS

👑 **Store Admin — `admin` / `admin123`**
Terminal: Admin Command Center | Scope: RBAC, live heartbeats, Daily Z-Report balancing, audit trail.

🩺 **Pharmacist — `pharmacist` / `pharma123`**
Terminal: Clinical Dispensing | Scope: eRx intake, CDSS allergy shield, Schedule X narcotic sign-off.

💳 **POS Cashier — `cashier` / `cash123`**
Terminal: Express Billing | Scope: 1-click eRx import, barcode scan, FEFO batch selection, UPI QR.

📦 **Inventory — `inventory` / `inv123`**
Terminal: Inwarding Bay | Scope: Multi-batch FEFO ledger, Add Medicine, Cold-chain IoT (2-8°C).

🛵 **Delivery Agent — `delivery` / `deliv123`**
Terminal: Logistics Dock | Scope: Kanban dispatch, cold-bag checklist, 4-digit OTP e-POD.

RUBRIC: LIVE APPLICATION WALKTHROUGH

Live Demo Walkthrough: Clinical CDSS & POS Billing

🩺 PHASE 1: CLINICAL PHARMACIST WALKTHROUGH

Step 1.1: Workstation Boot & Persona Select
Click '🩺 Clinical Pharmacist' tile on the login portal. System automatically loads station credentials (`pharmacist` / `pharma123`).

Step 1.2: e-Prescription Queue Review
Open Clinical Queue. Inspect incoming prescription `RX-2026-001` containing Amoxicillin for a patient with recorded Penicillin allergy.

Step 1.3: Trigger CDSS Allergy Interception (Poka-Yoke)
Click 'Run CDSS Safety Scan'. System immediately flashes a red contraindication banner: 'CRITICAL ALLERGY CONFLICT: Penicillin hypersensitivity'. Disables direct checkout.

Step 1.4: Schedule X Narcotic Double Sign-Off
Open narcotic prescription `RX-2026-004` (Zolpidem). Click 'Sign-Off'. Enter Doctor DMC No: `DMC-2024-8891` and Pharmacist PIN: `9944`. System verifies and logs to statutory `narcotic_register`.

💳 PHASE 2: POS CASHIER TERMINAL WALKTHROUGH

Step 2.1: Login to Front Counter A
Authenticate as Cashier (`cashier` / `cash123`). Workstation UI loads POS billing counter with barcode scanner focus.

Step 2.2: 1-Click Approved eRx Import
Click 'Import Approved eRx'. Approved prescription is pulled directly into the POS cart without manual re-typing, eliminating transcription error.

Step 2.3: Automated FEFO Batch Selection
Observe cart lines: system automatically selected the nearest-expiry batch (`days_to_expiry` lowest) to prevent dead-stock accumulation.

Step 2.4: Multi-Tender Payment & Thermal Receipt
Click 'Checkout'. Select Cash: enter ₹1000 for ₹740 bill -> automated change calculation: ₹260. Switch to Dynamic UPI QR -> simulate payment. Click 'Print Thermal Receipt' -> generates 80mm GST invoice.

RUBRIC: LIVE APPLICATION WALKTHROUGH

Live Demo: Inventory IoT, Delivery OTP & Z-Report

📦 PHASE 3: SUPPLY CHAIN & IOT ALARM

INVENTORY INWARDING & COLD-CHAIN

- Login as Inventory Manager (`inventory` / `inv123`).
- FEFO Ledger: Inspect batch list with color-coded expiry badges (<30d Red, <90d Yellow, >90d Green).
- Add Medicine & GRN: Inward new consignment, updating stock atomically.
- IoT Cold-Chain Monitor: Observe 2°C–8°C gauge. Click 'Simulate Temp Spike' (9.4°C) -> instant alarm badge & audit entry.

🛵 PHASE 4: DELIVERY FLEET & OTP

LOGISTICS FULFILMENT (E-POD)

- Login as Delivery Agent (`delivery` / `deliv123`).
- Kanban Dispatch Board: View orders under `Packaging`, `Out for Delivery`, `Delivered`.
- Cold-Bag Checklist: Confirm ice-gel insulation for insulin orders.
- 4-Digit OTP Handover: Customer provides OTP `1234`. Enter OTP -> system verifies and advances order to `DELIVERED`.

👑 PHASE 5: ADMIN PRESENCE & RBAC

STAFF GOVERNANCE & LIVE TELEMETRY

- Login as Store Admin (`admin` / `admin123`).
- Staff User CRUD: Add new staff account, edit roles, toggle status (`Active` / `Suspended`).
- Active Workstations: View live telemetry stream of connected terminals, IPs, and 5s heartbeats.
- 21 CFR Part 11 Audit Trail: Inspect tamper-evident operational event stream in real time.

💰 PHASE 6: DAILY Z-REPORT CASH CLOSE

FISCAL REGISTER BALANCING

- Open 'Daily Z-Report' register close module.
- System computes: `Expected Cash = Opening Float + Cash Sales - Payouts`.
- Enter counted physical cash: system evaluates live variance (`BALANCED ₹0.00` or `SHORTAGE`).
- Sign & Close Register: Locks register, logs to `z_reports`, and generates official printable Z-slip.

RUBRIC: IMPLEMENTED FUNCTIONALITY

Quality Verification & 24 Automated Test Results

24 / 24

Automated Tests Passed

100% pass rate across end-to-end unit, integration, and security test suites.

0%

Medication Dispensing Errors

Achieved via automated Poka-Yoke CDSS allergy and interaction barriers.

100%

Cold-Chain Integrity (2–8°C)

Continuous 30s background telemetry streaming with instant excursion alerts.

₹ 0.00

Cash Drawer Variance

Achieved on balanced registers via Daily Z-Report mathematical reconciliation.

📊 TRADITIONAL PHARMACY VS. AEGISPHARM TQM BENCHMARK

Quality Dimension	Traditional Retail Pharmacy (Manual)	AegisPharm Enterprise (TQM Driven)
Prescription Safety	Manual pharmacist check; vulnerable to handwriting errors	Automated CDSS cross-checks with patient allergy EHR
Inventory Picking	Random or FIFO; older batches expire on shelves	Algorithmic FEFO (First-Expiry, First-Out) guarantee
Cold-Chain Monitoring	Infrequent manual thermometer checks; silent spoilage	Continuous 30s IoT telemetry with automatic alarm alerts
Shift Cash Balancing	Paper ledger estimations; undetected shrinkage	Daily Z-Report mathematical reconciliation & signed Z-slip

REVIEW 1 SUMMARY & EXAMINER VIVA DEFENSE

Review 1 Rubric Criteria Fully Delivered

- **Q: Why SQLite instead of MySQL/MongoDB?**
  A: Pharmacy dispensing requires zero-downtime, local-first offline resilience. SQLite provides 100% ACID compliance with WAL mode, sub-10ms query commit latency, and zero server crashes, while FastAPI supports networked counter syncing.
- **Q: How does CDSS prevent medical malpractice?**
  A: Deterministic Poka-Yoke cross-referencing patient allergy arrays and drug-drug contraindication matrices, mathematically blocking checkout without supervisor override.
- **Q: How does FEFO reduce financial wastage?**
  A: Sorts batch ledger by earliest expiration (`days_to_expiry ASC`), ensuring nearest-expiry viable lots are dispensed first, with automatic lockout of expired stock.

🙏 THANK YOU RESPECTED FACULTY & EVALUATORS

Sincere gratitude to the Computer Science & Engineering faculty at Amrapali University for their academic guidance.

💬 **The floor is now open for questions, code inspection, and live application demonstration. (Candidate: Piyush Bhatt)**